BLDC48Controller

Float Golden Model - Mathematical Model Summary

2026-10-05 - Baseline for the Fixed-Point / FPGA Quantization Stage

This document records the continuous mathematical model confirmed and validated today. The Float Golden Model defines what the control mathematics should do before FPGA word lengths, LUT quantization, integer arithmetic, counters, and clock-cycle effects are introduced.

1. Design Principles

- The current stage is the Float Golden Model. Angles, amplitudes, and timing are represented using continuous physical quantities wherever possible.

- The legacy FPGA project is an important reference, but not an unquestionable specification. Sound concepts are retained; historical constants, obsolete scaling, and questionable logic are not copied blindly.

- Once accepted, the Float model becomes the reference truth. The later Fixed-Point Golden Model should approximate this model, not force the Float model to reproduce legacy RTL limitations.

2. Complete Signal Chain

Encoder count phi -> relative mechanical angle theta_m -> electrical angle theta_e -> voltage-vector angle theta_v -> phase advance -> theta_control -> 22 phase angles -> sinusoidal phase-voltage commands -> 22 duty ratios -> center-aligned PWM -> 22 complementary High/Low pairs -> physical deadtime -> 44 gate outputs.

3. Frozen Configuration Parameters

Parameter	Value	Meaning
LM10_COUNTS	4096	Encoder counts per mechanical revolution
POLE_PAIRS	3	Motor pole-pair count
NUM_PHASES	22	Number of motor phases
ENCODER_OFFSET	1022	Encoder / magnetic zero offset
ELECTRICAL_DIRECTION	-1	Electrical angle decreases with encoder count
TORQUE_DIRECTION	-1	Voltage vector uses the -90 deg direction
PHASE_ADVANCE_DEG	10.0 deg	Additional electrical phase advance
VOLTAGE_AMPLITUDE	0.5	Normalized sinusoidal modulation amplitude
PWM_FREQUENCY_HZ	15,000 Hz	Ideal Float PWM frequency
DEADTIME_LH_NS	40 ns	Low OFF -> High ON
DEADTIME_HL_NS	40 ns	High OFF -> Low ON
DTC_ENABLE	False	Dead-time compensation disabled
DEADTIME_COMPENSATION_NS	40 ns	Reserved interface; not used yet

4. Encoder Count to Mechanical Angle

The raw mechanical angle is:

theta_m,raw = phi / 4096 * 360 deg

The control calculation uses the position relative to ENCODER_OFFSET: relative_count = (phi - 1022) mod 4096, then theta_m = relative_count / 4096 * 360 deg. This separates the encoder mounting offset from the control-system zero.

5. Mechanical Angle to Electrical Angle

The motor has three pole pairs, so one mechanical revolution contains three electrical cycles. The confirmed direction is ELECTRICAL_DIRECTION = -1:

theta_e = (-1) * 3 * theta_m (mod 360 deg)

Thus 360 mechanical degrees correspond to 1080 electrical degrees. The full-revolution simulation confirmed exactly three electrical cycles.

6. Voltage-Vector Angle and Torque Direction

The simplified torque-producing stator voltage vector is positioned at +/-90 electrical degrees relative to the rotor electrical angle. The current direction setting is TORQUE_DIRECTION = -1:

theta_v = theta_e - 90 deg (mod 360 deg)

ELECTRICAL_DIRECTION and TORQUE_DIRECTION must remain separate: the former defines encoder-to-electrical-angle direction; the latter defines the stator-vector lead/lag direction.

7. Phase Advance

Phase advance compensates for phase lag associated with inductance, back-EMF, and current response. The current model uses PHASE_ADVANCE_DEG = 10 deg:

theta_control = theta_v + TORQUE_DIRECTION * PHASE_ADVANCE_DEG (mod 360 deg)

With the current signs, theta_control = theta_v - 10 deg. This is the physical-degree counterpart of the legacy PHASE_ADV input.

8. Spatial Distribution of the 22 Phases

The 22 phases are uniformly distributed over one electrical cycle:

Delta theta_phase = 360 deg / 22 = 16.363636... deg

For k = 0...21: theta_phase,k = (theta_control - k * 360 deg / 22) mod 360 deg

These are not 22 rotor electrical angles. They are 22 winding phase coordinates of the same commanded voltage vector. Phases separated by 11 positions are ideal opposites.

9. Sinusoidal Modulation and Voltage Amplitude

For each phase: s_k = sin(theta_phase,k)

The normalized phase-voltage command is: v_k = A * sin(theta_phase,k)

With A = VOLTAGE_AMPLITUDE = 0.5, v_k lies in [-0.5,+0.5]. This is the Float-model counterpart of legacy VS_RHO. The legacy signal is a 12-bit integer; the Float model first expresses the intended behavior as a continuous normalized amplitude.

10. Phase Voltage to PWM Duty

The bipolar normalized voltage command is mapped to [0,1] duty using:

D_k = (1 + v_k) / 2

With A = 0.5, the ideal duty range is 25% to 75%. A = 0 produces 50% on every phase; ideal A = 1 can span 0% to 100%.

The 4096-point full-revolution simulation confirmed approximately 25% to 75% globally. Opposite phases have duties summing to 100%, with numerical error around 4.44e-16.

11. PWM Form and Time Base

The Float model uses center-aligned PWM. At 50% duty the waveform is OFF from 0-25%, ON from 25-75%, and OFF from 75-100%.

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f_PWM = 15.000 kHz

T_PWM = 1 / 15000 = 66.6666667 us
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The legacy FPGA period is 13,328 clocks at 200 MHz, approximately 66.64 us / 15.006 kHz. The Float model intentionally uses the exact physical target; the clock quantization is deferred to the Fixed-Point stage.

12. Complementary High/Low Gate Signals

> Before deadtime, each phase produces High = PWM and Low = 1 - PWM. Twenty-two phases therefore produce 44 logical gate outputs.

The no-deadtime complementary-gate simulation has been validated and serves as the baseline for the deadtime stage.

13. Physical Gate Deadtime

Deadtime prevents simultaneous high-side and low-side conduction. In the Float model it is an exact physical time, not an FPGA clock count.

> Low -> High = 40 ns; High -> Low = 40 ns. Turn-off occurs immediately and the opposite turn-on is delayed by 40 ns.

For Low-to-High, Low turns OFF at the ideal rising edge and High turns ON 40 ns later. For High-to-Low, High turns OFF at the ideal falling edge and Low turns ON 40 ns later.

The 20,000 ordinary samples are only for plotting resolution. Exact switching-event times are inserted separately, so the 40 ns interval is not defined by the approximately 3.333 ns plotting sample period.

Both transition directions were verified at high zoom, and no High/Low simultaneous-ON state exists.

14. DTC - Dead-Time Compensation

DTC is separate from physical gate deadtime. Physical deadtime protects the devices; DTC attempts to compensate the average output-voltage error introduced by deadtime.

The legacy FPGA adds DTC above the PWM center and subtracts DTC at or below the center. The idea is meaningful, but this implementation is discontinuous around 50% duty.

> Example: with center = 3332 and DTC = 20, 3332 -> 3312 while 3333 -> 3353. A one-count input change causes a 41-count compensated jump.

The new model will not copy this algorithm. A more physical compensation method should use phase-current direction because winding current continues through freewheeling paths during deadtime.

> Current decision: DTC_ENABLE = False. DEADTIME_COMPENSATION_NS = 40 ns is only a reserved configuration interface and does not currently affect calculations.

15. Mapping to the Legacy FPGA Project

Legacy concept	New Float model	Decision
LM10 4096 counts	LM10_COUNTS = 4096	Retained
POLE_PAIRS = 3	POLE_PAIRS = 3	Retained
INIT_PHI = 1022	ENCODER_OFFSET = 1022	Physical meaning retained
Negative electrical direction	ELECTRICAL_DIRECTION = -1	Explicitly parameterized

Legacy concept	New Float model	Decision
12-bit VS_RHO	VOLTAGE_AMPLITUDE = 0.5	Normalized physical quantity first
PHASE_ADV	PHASE_ADVANCE_DEG	Directly expressed in degrees
22-phase 290/291 steps	Exact 360/22 spacing	Legacy quantization not copied
1600/3200 angle constants	90 deg / 180 deg	Obsolete 6400-count constants not copied
CLK_DIV = 3332	Not part of Float control math	Deferred to fixed-point/scheduling
PWM = 13328 clocks	15.000 kHz	Ideal physical specification first
DEADTIME_LH/HL = 8 clocks	40 ns / 40 ns	Physical time in Float model
DTC compare +/- offset	Interface retained; algorithm disabled	Discontinuous legacy logic not copied

16. Completed Simulation Validation

- At phi = 1024: raw mechanical = 90 deg; relative mechanical = 0.17578125 deg; electrical = 359.47265625 deg; voltage-vector angle = 269.47265625 deg; final control angle = 259.47265625 deg.

- Phase 1 voltage is approximately -0.491584, giving approximately 25.421% duty. Phases separated by 11 positions are opposite and their duties sum to 100%.

- A full 4096-count mechanical revolution produces exactly three electrical cycles.

- Center-aligned PWM uses the correct physical time axis and 15.000 kHz / 66.666667 us ideal period.

- High/Low are exact complements before deadtime.

- Both deadtime directions are exactly 40 ns, with no simultaneous High/Low conduction.

- The model has been extended to 22 phases / 44 High-Low gate outputs. Plotly initial-visibility behavior is only a UI issue and does not affect mathematical or CSV results.

17. Current Boundary of the Float Model

The current model describes encoder angle -> open-loop voltage-vector/sinusoidal modulation -> PWM/gate generation. It is not a complete motor electromagnetic simulation and is not a closed-loop FOC model.

- Motor R/L, back-EMF, inertia, and load dynamics are not modeled.

- The 22 phase-current feedback signals are not currently used.

- No Clarke/Park transforms or PI current loops are active.

- DTC is disabled.

- MOSFET/IGBT voltage drops, diode behavior, switching losses, and detailed power-stage effects are not modeled.

- These omissions do not prevent this model from serving as the Golden Model for the current modulation and gate-generation chain.

18. Next Stage: Fixed-Point / Quantization

The next stage will map the Float model into an FPGA-realizable integer model, comparing every stage against the Float reference.

- Angle representation: choose counts per revolution, word lengths, wrap behavior, and encoder-to-electrical-angle mapping.

- 22-phase spacing: quantize exact 360/22 without reintroducing unexplained 290/291 magic constants.

- Sine LUT: choose address width, LUT depth, signed output format, amplitude width, and quantify LUT error.

- Amplitude / VS_RHO: define the exact mapping between VOLTAGE_AMPLITUDE and the fixed-point amplitude representation.

- Multiplication/scaling: choose product widths, shifts, rounding versus truncation, and saturation behavior.

- Duty/PWM counter: map the 200 MHz clock to the 15 kHz target and quantify integer-period frequency error.

- Deadtime: at 200 MHz, 40 ns equals exactly 8 clock cycles.

- Final verification: Float -> Fixed-Point -> SystemVerilog, ensuring differences are intentional quantization rather than logic errors.

19. Frozen Baseline

> As of 2026-10-05, the accepted Float Golden Model chain is: Encoder -> electrical angle -> voltage-vector angle -> phase advance -> 22-phase sine generation -> normalized voltage -> duty -> center-aligned PWM -> complementary High/Low -> 40 ns physical deadtime.

During quantization, every fixed-point choice should document its word length, quantization step, rounding/truncation point, resulting error, and whether the design still satisfies the frozen properties: 22-phase symmetry, three electrical cycles per mechanical revolution, PWM-frequency target, and 40 ns deadtime.